Impedence Matching

In a transmission line, the maximum power transfer occurs, when the line is terminated with its characteristic impedence.

ie) Condition for maximum power transfer is that the load impedence must be equal to characteristic impedence. ie) $Z_L = Z_0$

A transmission line satisfying this condition is called matched line.

For a matched line, $\Gamma_R = 0$ & $S = 1$

ie) Reflected voltage is zero.

⇒ There is no reflection.

Practically transmission lines have impedence mismatch.

ie) $Z_L \neq Z_0$

Methods commonly employed for impedence matching are

(1) Transformer matching

(2) Lumped element matching

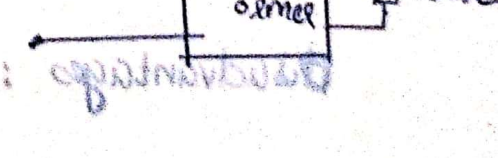
(3) Quarter wave transformer

(4) Stub matching.

(1) Transformer matching

By using a transformer with suitable turns ratio, impedence matching can be done.

Consider a transmission line having impedence 50Ω and load 100Ω. By using transformer with proper turns ratio as shown can be used to make load impedence impedence equal to characteristic impedence.



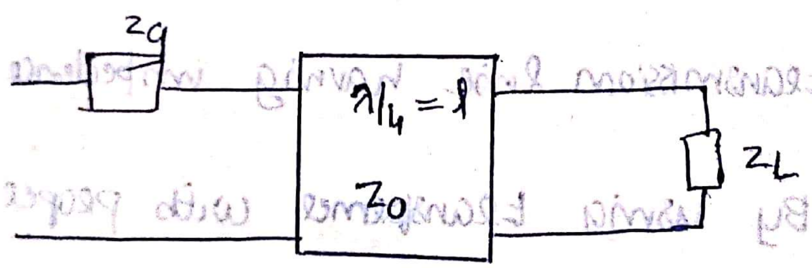
(2) Lumped element matching

A simple LC network can be used for impedence matching instead of transformer.

(3) Quarter wave transformer

A transmission line of length $\lambda/4$ is inserted in to the transmission line for impedance matching.

Two impedences can be matched by a section of transmission line which is quarter wave length long and has a characteristic impedence equal to geometer mean of load and generator impedence.



$$Z_0 = \sqrt{Z_g Z_L}$$

Disadvantages :

↳ It work only for a particular wavelength

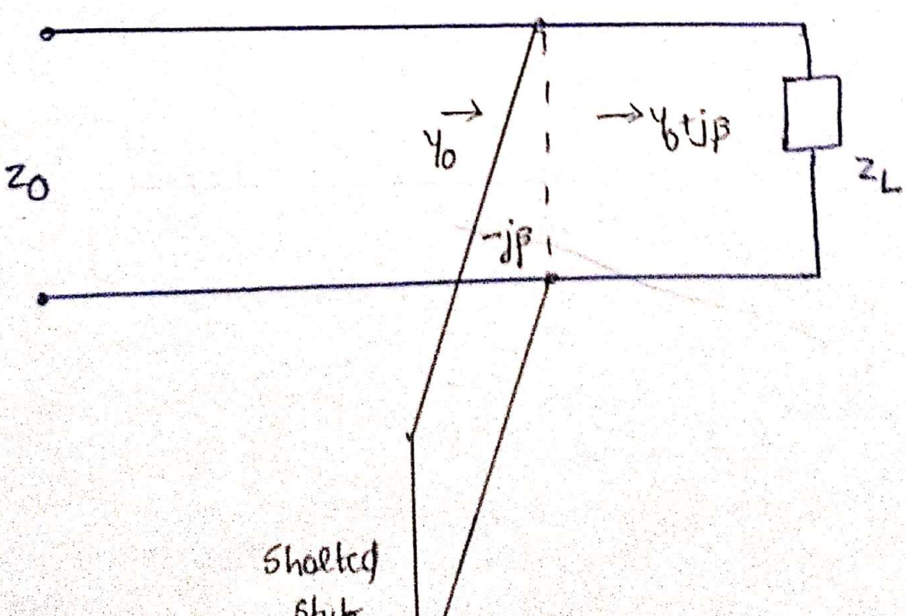
↳ Only for narrow band width

(4) ## Stub matching

Suitable sections of short circuited lines called stubs connected in parallel with the line at selective positions produce impedence matching.

Two types of stub matching → Single stub matching.
→ double stub matching

principle : A position on the transmission line nearest to the load is selected such that real part of line impedence at that point is equal to Z_0. Imaginary part is cancelled by the reactance of stub.



Advantages :

↳ superior to quarter wave transformer

↳ Applicable to all loads

Disadvantages :

↳ Cannot match variable impedence

↳ change in Z_L results in change in stub location.

But stub is mechanically fixed